

Project Title
Mini Search Engine / Information-Retrieval System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Mini Search Engine / Information-Retrieval System** is designed to index a document corpus and efficiently answer different types of user queries. The system supports four major query classes: **exact keyword search, phrase/substring search, fuzzy “Did You Mean?” search, and document-similarity search**. A suitable indexing structure, such as an **inverted index**, is implemented to enable fast keyword retrieval. Phrase and substring queries are processed using efficient string-matching techniques, while fuzzy search uses **edit distance** to identify and rank closely matching terms. Document similarity is determined using **term-frequency-based representations and similarity measures** to retrieve related documents. All core algorithms are **implemented from scratch**, without relying on built-in search-engine libraries, to demonstrate their underlying principles. The project also analyzes the **time and space complexity** of each algorithm and discusses appropriate choices for scalability. The system provides a practical demonstration of fundamental **Data Structures and Algorithms, Information Retrieval, indexing, ranking, and search optimization** concepts.

Signature of the Guide